MID-TERM EXAMINATION

(CBCS)

Course Name: B.Tech, Semester: 3rd

(Oct, 2022)

Subject Code: BCS 203 **Subject: Discrete Structures**

Time: 1 ½ Hours **Maximum Marks: 30**

Note: Q. 1 is compulsory. Attempt any two questions from the rest.

Q1 **(2.5*4)**

(a) Construct the truth table for the compound proposition
P -> ¬Q ^ ¬P <-> ¬Q

(b) Find whether $(p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p)$ is true or false.

(c) Find the mo of positive integers less than equal to 300 and divisible by 2 or 3

(d) Draw Venn diagrams for each of these combinations of the sets A, B, and C

i. A∩(BUC)
ii. A'∩B'∩C'
iii. (A-B)U(A-C)U(B-C)

Q2 **(5,5)**

(a) A college awarded 38 medals in football, 15 in basketball, and 20 in cricket. If these medals went to a total of 58 men and only three men got medals in all three sports. How many received medals in exactly two of the three sports?

(b) P and Q are two propositions. Which of the following logical expressions are equivalent?

(i) P ∨ ¬Q
(ii) ¬ (¬P ∧ Q)
(iii) (P ∧ Q) ∨ (P ∧ ¬Q) ∨ (¬P ∧ ¬Q)
(iv) (P ∧ Q) ∨ (P∧¬Q) ∨ (¬P ∧ Q)

Q3 **(4,6)**

(a)(i) Determine whether a relation R on set of all real numbers is reflexive, symmetric, antisymmetric, and/or transitive, where (x,y) ∈ R if and only if
(i) x+y=0, (ii) xy **≥ 0 (iii) x=2y (iv) x=1**

(ii) Explain partial ordered set and total ordered set using an example. Whether $(Z^+, |)$ is a partial order or not. Justify your answer. (| = division)

b) Determine the truth value of the following propositions, where p(x,y) : $y < x^2$, x and y are real numbers

i. ∀ (x) ∀ (y), P(x,y)
ii. ∀ (x) ∃ (y) P(x,y)
iii. ∃ (x) ∀ (y) P(x,y)
iv. ∃ (x) ∃ (y) P(x,y)

Q4 **(7,3)**

(a) Construct a Hasse Diagram of the poset ({1, 2, 3, 4, 6, 8, 12, 24}, |) and answer the following Questions:

i. Find the maximal elements.
ii. Find the minimal elements.
iii. Find the greatest element if it exists.
iv. Find the least element if it exists.
v. Find all upper bounds of {1,2,3}
vi. Find all lower bounds of {1,2,3}

P.T.O.

vii. Find the least upper bound of {1,2,3} if it exists.

viii. Find the greatest upper bound of {1,2,3} if it exists.

(b)(i) Build arguments using rules of inference "Sandy works hard", "If Sandy works hard, then he is a dull boy", and "If Sandy is a dull boy, then he will not get the job". Therefore, "Sandy will not get the job".

(ii) Justify that if $\forall x(P(x) \rightarrow Q(x))$ and $\forall x(Q(x) \rightarrow R(x))$ are true, then $\forall x(P(x) \rightarrow R(x))$ is true, where the domain of all quantifiers are same.

$5\frac{1}{2}0$